

----- Assignment 1 -----

Code 1

```
member = int(input("enter the number of member:"))
books_barrow= [] * member
total_no = 0
zero_byer = 0
most_freq_byer = 0
# Adding number of book barrowed by member to an list
for i in range (member):
    print("Enter number of books borrowed by member ",i + 1)
    num = int(input("Enter:" ))
    books_barrow.append(num)
    i += 1
print("total books barrow by member :",books_barrow)

#Total of boks barrow
for i in books_barrow:
    total_no = total_no + i
    i += 1
print("total_books :",total_no)
# 1 calculating the avg of the book
avg_no =int( total_no/ member )
print( "the average no of book :",avg_no)

# calculating zero book member
highest= books_barrow[0]
lowest = books_barrow[0]
for i in range (1,member):
    if books_barrow[i] > highest:
        highest = books_barrow[i]
    if books_barrow[i]< lowest:
        lowest= books_barrow[i]

print("the higest buyer is :", highest)
print("the lowest buyer is :", lowest)

for i in books_barrow:
    if(books_barrow[num] == 0):
        zero_byer+=1
print("member of member who buy zero book ",zero_byer)

#calc most frequent buyer
for i in range(len(books_barrow)):
```

```
count = 0
for j in range(len(books_barrow)):
    if books_barrow[i]==books_barrow[j]:
        count+=1

print ("the most frequent books buyers :", count )
```

output :

enter the number of member:5

Enter number of books borrowed by member 1

Enter:8

Enter number of books borrowed by member 2

Enter:8

Enter number of books borrowed by member 3

Enter:4

Enter number of books borrowed by member 4

Enter:5

Enter number of books borrowed by member 5

Enter:0

total books barrow by member : [8, 8, 4, 5, 0]

total_books : 25

the average no of book : 5

the highest buyer is : 8

the lowest buyer is : 0

member of member who buy zero book 0

the most frequent books buyers : 1

----- Assignment 2 -----

Code :

```
list = [100, 102 ,103 ,106,108,109,110,111,113,114]
custId = int(input("enter the cust Id:"))

found = False
#===== Liner sarch
=====
print("Linear sarch:")
for i in range(len(list)):
    if(list[i]==custId):
        print("id matched")
        print(list[i])
        found = True
        break
if( not found):
    print("id not matched ")

#===== Binary sarch
=====
print("binary Sarch")
low = 0
high = len(list) - 1
found = False

while low <= high:
    mid = (low + high)//2

    if list[mid]==custId:
        print("Id found",list[mid])
        found = True
        break
    elif(custId < list[mid]):
        high = mid -1

    else:
        low = mid + 1
```

```
if not found:  
    print("element not found:")
```

output:

enter the cust Id:103

Linear sarch:

id matched

103

binary Sarch

Id found 103

----- Assignment 3 -----

Code:

```
# Display highest sal using bouble sort

sal = []
n = int(input("enter the number of staff"))
for i in range(n):
    sal.append(int(input("Enter the sal of staff ")))

for i in range(n-1 ):
    for j in range(n-1 - i):
        if sal[j]> sal[j+1]:
            sal[j],sal[j+1]= sal[j+1],sal[j]
print(sal)

#Display highest sal using selection sort
sallary = []
num = int(input('enter the number of staff'))
for i in range(num):
    sallary.append(int(input("Enter the sal of staff ")))
count = len(sallary)

for i in range (count - 1 ):
    min_index = i

    for j in range(i + 1,count ):
        if sallary[j] < sallary[min_index]:
            min_index = j

    sallary[i], sallary[min_index]= sallary[min_index],sallary[i]
print(sallary)
```

Output:

enter the number of staff5

Enter the sal of staff 20000

Enter the sal of staff 25000

Enter the sal of staff 38000

Enter the sal of staff 45000

Enter the sal of staff 90000

[20000, 25000, 38000, 45000, 90000]

enter the number of staff5

Enter the sal of staff 20000

Enter the sal of staff 40000

Enter the sal of staff 30000

Enter the sal of staff 38000

Enter the sal of staff 44000

[20000, 30000, 38000, 40000, 44000]